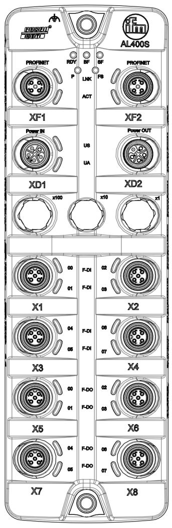
209_SRIO_CompSpec_Module_Com

System Component Specification Module Communication



Project: 153829 Remote_IO_Safe_DIDO_PROFIsafe

Affected Products: eco310 / **AL400S** / PROFIsafe 6x2 F-DI 2x2 F-DO IP67
eco310 / **AL401S** / PROFIsafe 6x2 F-DI 2x2 F-DO IP69K

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Approved Versions

Current Document version **V1.01** has **not** been approved.

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- **V1.00** (2026-06-25 05:36)

Reviewers for current version

Info: This document is reviewed by using tool-based review process. Every workitem is reviewed in the document context by the R&D experts independently. Therefore the workitems are checked for correctness, completeness and consistence to other work-items and to the document context. The minimal number of independent reviewers are defined by the inno-process.

➪ There are workItems which contains at least one waiting approval

Change Log

(Typ: e = draft , m = modified, + = added, - = deleted)

V1.00

Type	Chapter	Author	Description
c	all	B.Ruess	initial creation, added content and created first version

V1.01

Type	Chapter	Author	Description
m	5	B.Ruess	adapted req based on test feedback. See ~~SRIO-22447~~ - IoTCore Requirements updated

1 Scope

1.1 Purpose of this Document

This document describes the component COM.

1.2 Referenced Documents

REF	Document ID	Document name	Version
[1]	100_SRIO_Product-Requirements	Product Requirement Specification	V1.20
[2]	130_Communication	Product Requirement Specification: Communication	V1.18
[3]	140_FW-Update	Product Requirement Specification: Firmware Updates	V1.16
[4]	180_Housing	Product Requirement Specification: Housing	V1.16
[5]	190_QualityRequirements	Product Requirement Specification: Quality Requirements	V1.11
[6]	SafetyManual_STM32F7_V7	UM2318 STM32F7 series safety manual	V7
[7]	PROFIsafe Specification	PROFIsafe_3192_V26MU1_Aug18	V2.6 MU1 Aug18
[8]	ZVEI_24V_Schnittstellen_final.pdf	ZVEI_Positionpaper	2021.01 V1
[9]	VHIP_BuildingBlocks_Device_Information_r1081163.pdf	VHIP_BuildingBlock_Device_Information	V2.0
[10]	svn/.../ReactionTime	Folder contains Siemens .xls [e.g. s7safety_rtt_de.xls]	V12.3.0.0_0.0
[11]	FMEA for STM32F7 AN5137.pdf	FMEA of STM	Rev 3 (Jul - 2019)
[12]	FMEDA for STM32F7 20AN5132.pdf	FMEDA of STM	Rev 3 (Jul - 2019)
[13]	Safety Manual STL STM32F7	STM32F7 series self-test library safety manual	Rev 7 (Oct - 2023)
[14]	STL_F7_UG_cx630716_7.0.pdf	STM32F7 series self-test library user guide	Rev 7 (Aug - 2023)

1.3 Applicable standards

Standard	Title
IEC 61508 Part 1-7 :2010	*Functional safety of E/E/PE safety-related systems*
EN ISO 13849-1:2023	*Safety of machinery - Safety-related parts of control systems*
DIN EN 60068-2-6 and DIN EN 60068-2-64	*Environmental testing*
EN IEC 61000-6-2: 2019 EN IEC 61000-6-4: 2019	*Electromagnetic compatibility (EMC)*
EN 61010-1 regarding fire spread EN 61010-1 regarding clearance & creepage distance	*Safety requirements for electrical equipment for measurement, control, and laboratory use*

EN 61010-1 regarding resistance of the identification plate	
EN 61326-1: 2021 regarding the EMC EN 61326-3-1:2017 regarding the EMC (safety related levels)	*Electrical equipment for measurement, control and laboratory use – EMC requirements*
IEC 61131-2:2017 regarding the EMC resistance and mechanical robustness	*Programmable controllers*

1.4 Applicable documents

Nr.	Version	Title	Short Description
[SysArch-01]	SRIO-18382 - V2.02	201_SRIO_System_Spec_Architecture	Details the system architecture of SRIO
[SC-04]	SRIO-19197 - V2.02	204_Technical_Safety_Concept	Safety Concept of SRIO
[SysSCPU-05]	SRIO-18952 - V1.01	205_SRIO_CompSpec_Module_SCPU	Details the Component SCPU
[SysDI-06]	SRIO-19198 - V1.02	206_SRIO_CompSpec_Module_DI	Details the Component DI
[SysDO-07]	SRIO-19199 - V1.01	207_SRIO_CompSpec_Module_DO	Details the Component DO
[SysPS-08]	SRIO-18957 - V1.02	208_SRIO_CompSpec_Module_PS	Details the Component PS
[SysCOM-09]	SRIO-7664 - V1.00	209_SRIO_CompSpec_Module_Com	Details the Component COM
[Glossary]	SRIO-7218 - V1.00	297_Glossary_and_ListOfAbbreviations	Used terms and abbreviations in the project.
[Prd_Data]	SRIO-11112 - V1.00	220_ProductionData	Information regarding production relevant data
[Com_Spec]	SRIO-7664 - V1.00	209_SRIO_CompSpec_Module_Com	Details the Component COM

2 Internal Structure

2.1 VHIP 3 core

Hardware
The VHIP3 core consists of an STM32F777NI CPU from the manufacturer ST Microcontroller. The µC has 2MB Flash for instructions and 768kByte RAM for data onboard. An EEPROM for persistent data is connected to CPU. The core provides an external Flash for updates.

Characteristic	Value
CPU	Cortex-M7
external Flash	32 Mbit
EEPROM	128 Kbit

Table 1 : VHIP3 core hardware characteristics

Software
The VHIP3 software support contains build tools, CI components, example projects, scripts for production, release notes generator and OS components. The template project with µC/OS will be the entry point for software development. The building blocks mentioned in the detailed block diagram will be integrated and connected via inter process communication.

2.2 BB Fieldbus

The interface description is done in the chapter SRIO-9521 - Fieldbus of [SysArch-01] It contains details from the perspective of the device. This chapter should explain the view of the device with a focus on the configuration of building block.
The mapping for data of fieldbus is described in the following chapters e.g.
The configuration of BB-Fieldbus is explained along the numbers shown in the diagram.

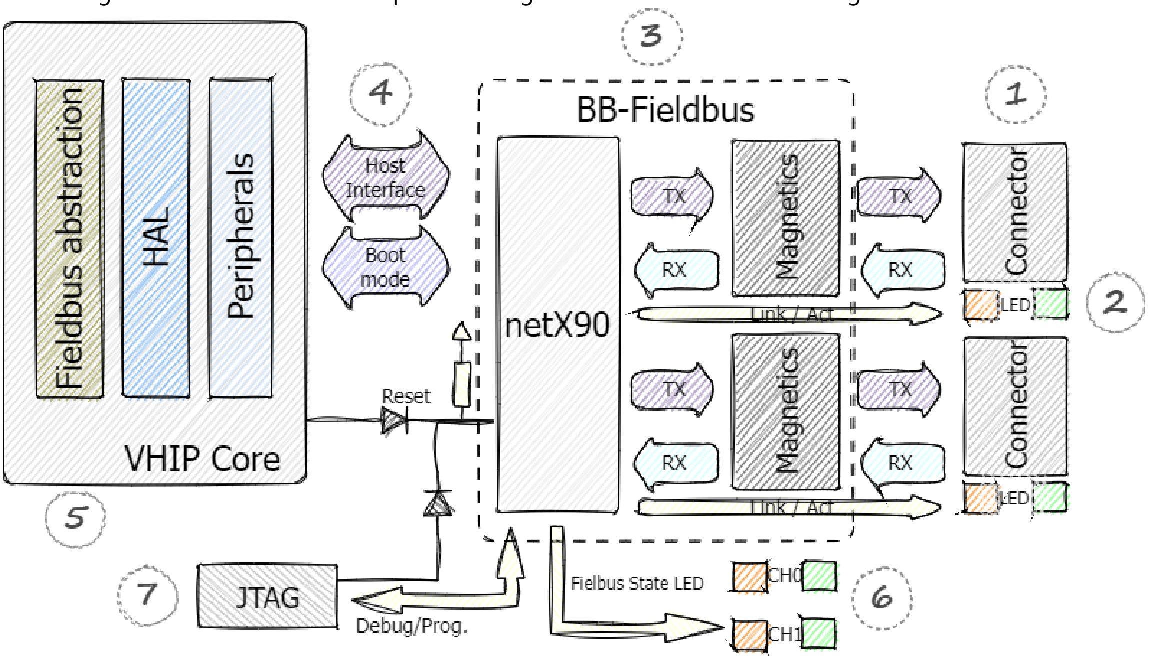


Figure 1 : BB Fieldbus configuration diagram

No.	description	configuration value
1	Connector	SRIO-711 - Two Fieldbus Connectors SRIO-713 - Fieldbus Connector
2	LED Ethernet	SRIO-8629 - LED Ethernet control SRIO-6526 - LNK LEDs SRIO-6525 - ACT LEDs
3	BB-Fieldbus	The building block contains netX90, clock, magnetics, etc. Hardware description: http://dettsvn.tt.de.ifm/svn/ifm_platform/INTERFACE/FBUS_ETH_100M_2P/NETX90/General/Description

		_Documents/NETX90_INT_FBUS_ETH_100M_2P_Module_Description.pdf
4	Host interface + Boot mode	SRIO-8633 - VHIP3 - BB Fieldbus serial SRIO-8632 - VHIP3 - BB Fieldbus DIRQ SRIO-8631 - VHIP3 - BB Fieldbus boot select
5	VHIP core	SRIO-1529 - PROFINET - Fieldbus Protocol PROFINET
6	FB Status LED	SRIO-8607 - Connection for BF and SF LED SRIO-6530 - BF LED SRIO-6529 - SF LED
7	Debug/Prog.	SRIO-8628 - Com Debug

2.3 Communication Module Software – Architecture Specification

2.3.1 Overview

The **Communication Module Software** acts as the central integration layer between external communication components and the internal system. Its primary responsibility is to provide a **uniform interface for data exchange and service operations**, ensuring real-time capability where required and abstracting product-specific details.

2.3.2 External Components

An External Component refers to any third-party or native module that is integrated into the system using its own proprietary interface. These components typically provide specialized functionality, such as communication protocols or hardware control, and are essential for extending the capabilities of the Communication Module Software without reinventing existing solutions. External components operate independently and maintain their own internal logic and lifecycle. They are not modified by the Communication Module Software; instead, they are accessed through their native APIs, which may include vendor-specific commands, configuration options, and data exchange mechanisms.

Included External Components:

- BB Fieldbus: A communication protocol stack used for industrial fieldbus networks.
- IOT Core
- NVMEM
- LED Control Module: A hardware-specific component responsible for managing LED indicators or lighting systems.

2.3.3 Interfaces

The Communication Module Software defines two primary types of interfaces to facilitate interaction between the internal system and external components: **Data Interfaces** and **Service Interfaces**. These interfaces differ in their purpose, timing requirements, and communication patterns.

2.3.3.1 Data Interface

The Data Interface is designed for real-time communication and is responsible for transferring operational data between the system and external components in a deterministic manner. It ensures that time-critical information, such as sensor readings or control signals, is exchanged without delays that could impact system performance.

Characteristics:

- **Real-Time Capability:** The interface guarantees compliance with strict timing constraints required for industrial and embedded applications.

- **Synchronous Call Behavior:** Data exchange occurs through synchronous function calls, ensuring immediate execution and predictable response times.
- **Deterministic Design:** The interface is optimized for consistent latency and minimal jitter, which is essential for real-time control loops.

2.3.3.2 Service Interface

The Service Interface provides access to non-real-time operations, such as configuration, diagnostics, and maintenance tasks. These operations do not require deterministic timing and can be executed asynchronously without affecting the core control processes.

Characteristics:

- **Asynchronous Call Behavior:** Service requests are typically handled in a non-blocking manner, allowing the system to continue its real-time operations while processing service tasks in parallel.
- **Flexible Execution:** While primarily asynchronous, the Service Interface may include certain synchronous operations for specific scenarios (e.g., immediate status queries or critical configuration changes). These details will be defined during the implementation phase.

Implementation Approach:
The Service Interface is implemented using **eRPC (Embedded Remote Procedure Call)** as the core mechanism for handling asynchronous service operations. eRPC provides a lightweight and efficient RPC framework tailored for **embedded environments**, particularly **multichip systems and heterogeneous multicore SoCs**. This makes it an ideal choice for the Communication Module Software, where tight coupling and low-latency communication are critical

Service Generation Using erpcgen:
To simplify integration and reduce manual coding, eRPC includes a **code generator tool called `erpcgen`**. This tool:

- Accepts **IDL (Interface Definition Language)** files with the `.erpc` extension.
- These IDL files define **data types** and **remote interfaces** for the services.
- Generates the **shim code** that handles serialization and invocation automatically.
- Output in **C++**.

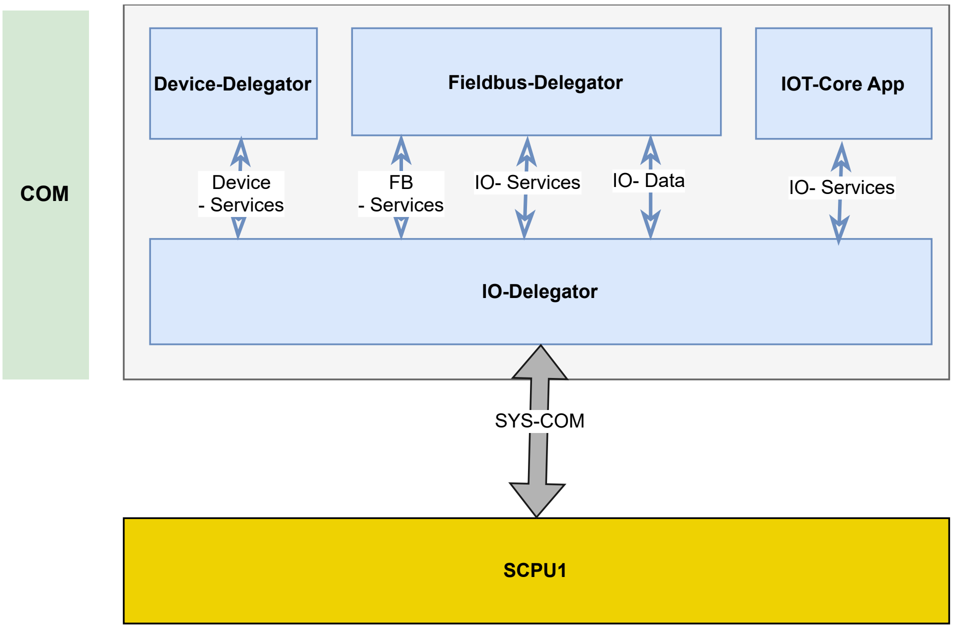
2.3.4 Delegator

The Delegator is a core architectural component that acts as an intermediary layer between the Communication Module Software and external components. Its primary purpose is to abstract the complexity of different implementations and provide a consistent and unified interface to the core application. This design ensures a clear separation of concerns and enhances system maintainability, scalability, and robustness.

One of the key responsibilities of the Delegator is **integration and isolation**. External components, including their proprietary logic and specific interfaces, are encapsulated within the Delegator. This prevents direct interaction between external systems and the internal architecture, thereby reducing coupling and shielding the system from potential changes in external implementations.
In addition, the Delegator provides **uniform interfaces** to the Communication Module Software. These include standardized data interfaces and service interfaces that follow a common design pattern. This approach allows the application layer to interact with different external components in a consistent manner, regardless of the underlying communication protocol or technology.
Despite offering a uniform interface, the Delegator also supports **product-specific adaptations**. This includes handling specialized mappings such as fieldbus-specific data structures, protocol conversions, or configuration transformations. By managing these adaptations internally, the Delegator ensures compatibility across different product variants while preserving the overall architectural consistency.

2.3.4.1 IO-Delegator

The IO-Delegator is responsible for the exchange of process data and includes the transport layer for SysCom communication between the Host CPU and Safe CPU1. The IO-Delegator operates with a cyclic task running at a cycle time of 2 ms.
It manages the continuous data exchange with Safe CPU1, including system parameters such as CPU temperature and current consumption. These values are received cyclically, stored locally, and updated upon each new data reception. The IO-Core accesses these values via service interfaces.



The SysCom transport layer implements a timeout mechanism for SPI communication. In cases where communication with the Safe CPU is interrupted or the Safe CPUs are not reachable, the system detects the failure. Consequently, the host application generates a diagnostic message to the PLC and logs the event in the system log buffer

2.3.4.2 Fieldbus-Delegator

The Fieldbus Delegator is responsible for communication with the fieldbus building block and implements all application-specific functionalities related to fieldbus integration.
This component operates with two tasks:

- A **cyclic task** with a cycle time of 2 ms
- An **acyclic task** with a cycle time of 5 ms

For process data exchange with the IO-Delegator, a shared memory mechanism is used. This ensures efficient and low-latency communication between components without requiring additional protocol overhead.

2.3.4.3 Device-Delegator

The Device Delegator manages all device-specific parameters required by both the fieldbus and IoT core functionalities.
During system startup, the IO-Delegator retrieves remanent (persistent) data stored in EEPROM and synchronizes it via the NVMEM Delegator. This ensures that all required configuration and state data are restored correctly after system initialization.
The Device Delegator operates with a cycle time of 10 ms and provides processed device parameters to other system components as needed.

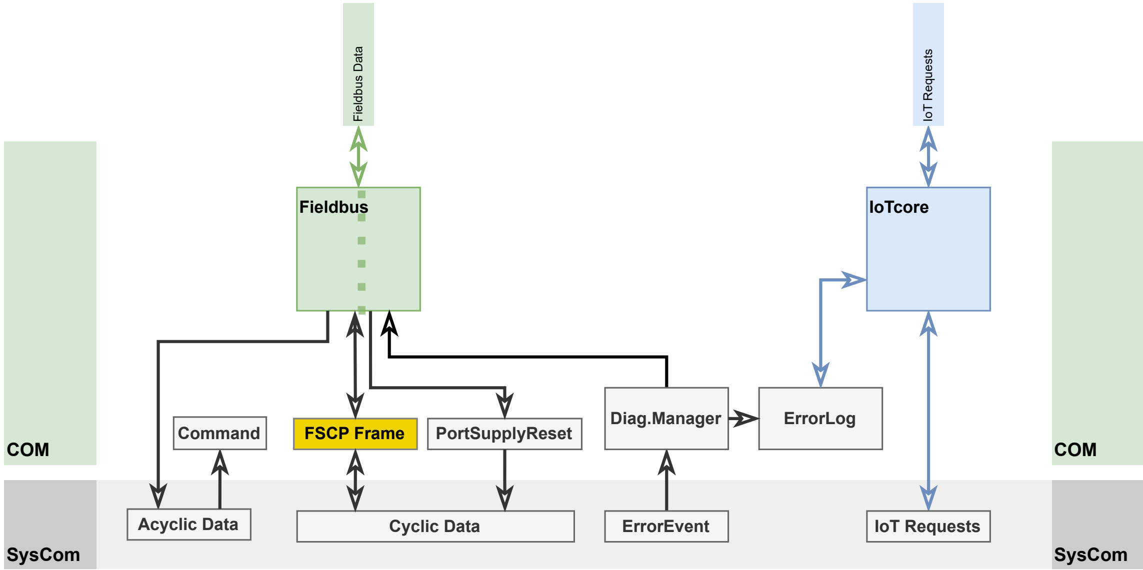
2.3.4.4 NVMEM-Delegator

The NVMEM Delegator is responsible for accessing non-volatile memory (EEPROM). Its main function is to read stored data and provide these values as services to the Device Delegator.
By centralizing access to persistent storage, the NVMEM Delegator ensures consistency, reliability, and controlled data handling. It also supports system recovery by enabling the restoration of saved system states after power cycles or restarts.

2.3.5 IOT-Core

3 Interfaces

COM has the external interfaces fieldbus (depends on Module Variant, see [SysArch-01 here: SRIO-9886 - Module Variants]) and obligatory IoTCore interface as well as the internal interface SysCom to the safe core (SCPU) and additional interfaces to BB-Fieldbus or Debug interfaces.



3.1 Fieldbus

3.1.1 AL40xS

As defined in [SysArch-01 here: SRIO-9886 - Module Variants] the AL40xS are PROFINET Fieldbus devices.
Therefore COM has the external fieldbus interfaces PROFINET additional to the obligatory IoTCore interface and the internal interface SysCom to the safe core (SCPU).

see below in chapter 6 - Profinet

3.2 IoTCore

see below in chapter 5 - IoTCore

3.3 SysCom

Verify in 201_SRIO_System_Spec_Architecture : -> SRIO-11276 - System Communication [SysCom]

3.4 Additional Internal Interfaces

Debug / Prog.

SRIO-8628 - Com Debug
A debug interface as a connector or contact plane must be on PCB. — Approved, Software, QM

SRIO-17163 - Com Debug Reset
The debugger must be able to reset the BB-Fieldbus without influencing the VHIP core. — Approved, Software, QM

SRIO-17162 - VHIP3 core Debug
The VHIP core must be able to reset the BB-Fieldbus without influencing the debugger. — Approved, Software, QM

Host Interface

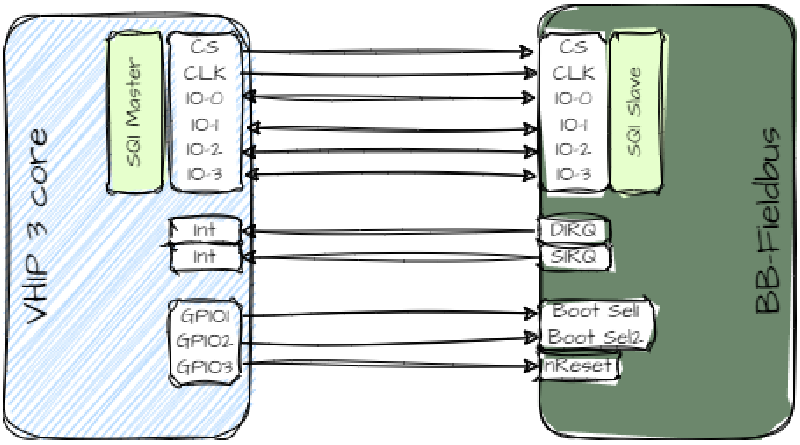


Figure 2 :VHIP3 core to BB-Fieldbus

SRIO-8633 - VHIP3 - BB Fieldbus serial

The connection between VHIP3 Core CPU and BB-Fieldbus must be done via QSPI. Approved, Hardware, QM

SRIO-8632 - VHIP3 - BB Fieldbus DIRQ

An interrupt input is to be connected to DIRQ and SIRQ of BB-Fieldbus. Approved, Hardware, QM

SRIO-8631 - VHIP3 - BB Fieldbus boot select

The boot select pins of BB-Fieldbus must be connected to the outputs of VHIP3 core. Approved, Hardware, QM

4 Functions

4.1 Uptime

SRIO-15218 - Errorhandling - Uptime - Timestamp

The uptime shall be counted as seconds and saved as 4 Byte Value. Approved, Software, QM

SRIO-15398 - Errorhandling - Uptime - Persistence

The uptime will be cleared/reset by coldstart. Approved, Software, QM

4.2 ErrorHandling

Describe behavior

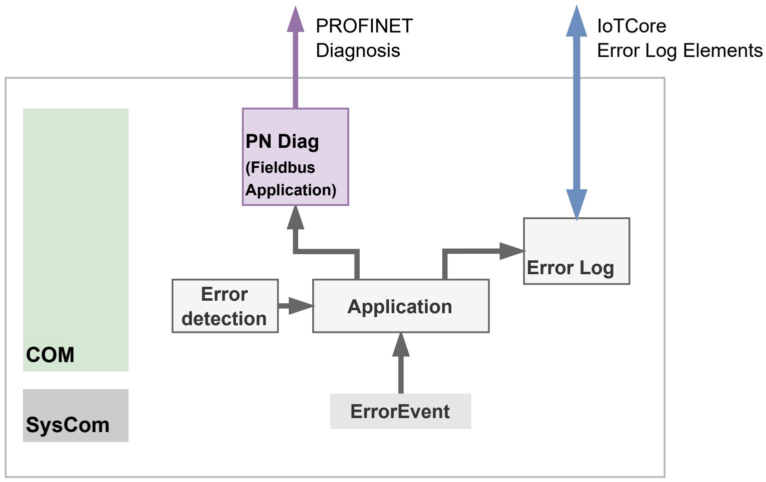


Figure 3 ErrorHandling

SRIO-15223 - Errorhandling - Error Detection

In case of an self detected Error, COM shall create an ErrorEvent using the same format as the ErrorEvents recieved by SCPU.
see SRIO-11026 - SysCom Channel: ErrorEvent
Note: Not applicable Data shall be set to "0".
Rational 1: "self detected" in the meaning of e.g. "Loss of Communication with SCPU".
Rational 2: COM, being stated as "QM" doesn't have to perform self tests on its own.

Approved, Software, PQS, QM

SRIO-15405 - Errorhandling - Error Information send via fieldbus

Each Error Event (self detected or recieved) shall be send to the PLC using the fieldbus diagnosis.

Approved, Software, PQS, QM

SRIO-10250 - Errorhandling - Fieldbus Diagnosis

The ErrorEvent send via fieldbus diagnosis shall be mapped to the respective fieldbus format.
See 6.6 - Diagnosis and Process alarms Approved, Software, PQS, QM

SRIO-10932 - Errorhandling - Fieldbus Diagnosis Event Mode

The distinction in the ErrorHeader[Mode] (Event appear/disappear) shall have an appropriate reaction (appear/disappear) for the fieldbus diagnosis.
Note: See SRIO-11026 - SysCom Channel: ErrorEvent and SRIO-12356 - SysCom - ErrorEvent: ErrorHeader Coding: Mode Approved, Software, PQS, QM

4.2.1 Error Logs

SRIO-15406 - Errorhandling - Error Information logged

Each Error Event (self detected or recieved) shall be written to the Error Log. Approved, Software, PQS, QM

SRIO-10703 - Errorhandling - Error Log - Entries Count

The ErrorLog shall contain at least 100 ErrorEvents. Approved, Software, PQS, QM

SRIO-10663 - Errorhandling - Error Log - Circular Buffer
The ErrorLog shall be designed as a Circular-Buffer. Approved, Software, PQS, QM

SRIO-6521 - Errorhandling - Error Log - Persistence
The Error Log shall be cleared by coldstart. Approved, Software, PQS, QM

SRIO-10662 - Errorhandling - Error Log - IoT Access
The Content in the Error Log shall be able readable via IoT. Approved, Software, PQS, QM

SRIO-15221 - Errorhandling - Error Log - IoT Elementcontent Readability
The Error Log content (ErrorEvents + Uptime) shall be understandable for the user.
Note: HEX Codes + Table in manual is counted as understandable. Approved, Software, PQS, QM

4.3 General

SRIO-21357 - SCPU Timeout
In case COM detects a timeout in the communication to SCPU, an error event shall be triggered by COM.

Rational: This error event improves the user experience to understand the root cause since the safe side is no longer reachable handles all other error events.
Parallel to this information, the timeout will also be handled by the safe communication on the SCPU side as well as in the plc (black channel principle). Approved, Software, QM

5 IoTCore

5.1 General

SRIO-7913 - IoT-Core API and Catalogue
The IoT-Core shall support IoT-Core API V2.0 and Catalogue V2.0 Approved, Software, QM

5.2 Features

SRIO-7914 - IoT-Core Visualizer
The IoT-Core Visualizer compact shall be part of the device and shall be accessable by IP-Adress.
Approved, Software, PQS, QM

SRIO-7912 - IoT-Core Visualizer - favicon
A favicon should be shown in browser when opening the IoTCore visualizer. Approved, Software, QM

SRIO-7903 - network adapter - HTTP
The device shall support HTTP Approved, Software, QM

SRIO-7907 - network adapter - HTTPS
The device shall not support HTTPS. Approved, Software, QM

SRIO-7906 - network adapter - Websocket
The device shall not support Websocket. Approved, Software, QM

SRIO-7905 - network adapter - MQTT
The device shall not support MQTT. Approved, Software, QM

SRIO-7908 - Connection Usecase
The device shall support only incoming connections and no outgoing connections.
Remark: Outgoing connections e.g. to push information's (ifm "subscriptions functionality") are not supported.
Approved, Software, QM

SRIO-7909 - Number of incoming http connections
The number of active http connections shall be 2. Approved, Software, QM

SRIO-7915 - Software update
The device shall receive and install an software update over the IoT-Core. Approved, Software, QM

5.3 IoTCore Definitions

5.3.1 Tree Element Mapping

Table 2 Glossary of following tables [definitions are synced with other projects, some definitions are not used for this product]

Rigths/ Usage	Definition
COM	COM-Module, CPU3
SCPU	SCPU-Module, CPU1
R	Read
W	Write
SB	Only possible to write this values via a setblock
S	Service
[U]	Device has to be in State Update
(H)	Hidden in tree
(P)	Only in production mode (= production_fuse not set)

[DO]	Only in DI/DO
[PLC]	Requires disconnected PLC to set data

5.3.1.1 General elements

SRIO-2938 - The element tree of the Safe Remote IO is available in following structure:

Tree element	Source of value	Rights	Format (example)
Identifier of root node	Needs to be "unique" "[vendor]-[devicename]-[serialnumber]"	R	ifm-AL400S-26415630044
/deviceinfo	defined in 5.3.1.2 - Deviceinfo	R	-
/devicetag	defined in 5.3.1.3 - Devicetag	RW	-
/devicestatus	defined in 5.3.1.4 - Devicestatus	R	-
/devicecontrol	defined in 4.2.4 - devicecontrol	RW	-
/systemtime	defined in 5.3.1.6 - Systemtime	RO	-
/io/di	defined in 4.2.6 - DI	R	-
/io/do	defined in 4.2.7 - DO	R	-
/fieldbussetup	defined in 5.3.1.9 - Fieldbussetup	R/RW	-
/safecom	defined in 5.3.1.10 - SafeCom elements	R	-
/firmware	defined in 5.3.1.11 - Firmware	R/W/S	-
/fit	defined in 5.3.1.12 - FIT	R/(H)/(P)	

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5.3.1.2 Deviceinfo

SRIO-2946 - Deviceinfo provides the following elements:

tree element	Source of value	Rights / View	Profile	ElementType / DataType	Format (examples)
/deviceinfo		R	deviceinfo	structure	IOTC-600 - The deviceinfo profile
/deviceinfo/productcode	COM	R		data / String	e.g. 'AL400S'
/deviceinfo/productname	COM	R		data / String	e.g. 'ETH Module PROFIsafe 6x2 F-DI 2x2 F-DO IP67'
/deviceinfo/vendor	COM	R		data / String	'ifm electronic gmbh'
/deviceinfo/devicefamily	COM	R		data / String	'srio-pn-ps'
/deviceinfo/serialnumber	COM	R		data / String	'26415630044'
/deviceinfo/productiondate	COM	R		data / String	'15.01.2026, 10:55:03'
/deviceinfo/hwrevision	COM	R		data / String	'AA'
/deviceinfo/hwversion	COM	R		data / String	'1.4.0.1'
/deviceinfo/swrevision	COM	R		data / String	Version Update Container

					e.g. '1.0.0.0'
/deviceinfo/swinfo				Structure	
/deviceinfo/swinfo/cpubootloaderversion	COM	R		data / String	e.g. 'VH1xxx_bl_4.1.1.357'
/deviceinfo/swinfo/scpuversion	SCPU	R		data / String	e.g. '0.0.20.4364'
/deviceinfo/swinfo/scpubootloaderversion	SCPU	R		data / String	e.g. '1.0.2.29'
/deviceinfo/fieldbustype	COM	R		data / enum	0 = Profinet

Ready for Review, Software, User-Documentation, QM

5.3.1.3 Devicetag

SRIO-2955 - Devicetag provides the following element:

tree element	Source of value	Rights / View	Profile	ElementType / DataType	Format (examples)
/devicetag		R	devicetag	structure	
/devicetag/applicationtag	COM	RW		data / String (maxlength: 31)	"factoryXYZ_machine123"

Approved, Software, QM

5.3.1.4 Devicestatus

SRIO-2950 - Devicestatus provides the following elements:

tree element	Source of value	Rights / View	Profile	ElementType / DataType	Format (examples)
/devicestatus			devicestatus	Structure	IOTC-859 - The devicestatus profile
/devicestatus/operatingstate	SCPU	R		data / Enum	1: Initialization 2: Update 3: Parametrization 4: Operate 5: Stop
/devicestatus/operatinghours	COM	R		data / Number	Feature is not supported. Return status code: 501
/devicestatus/rotary_switch	SCPU	R		data / Number	123
/devicestatus/temperature/scpu1	SCPU	R		data / Number	52 (°C)
/devicestatus/temperature/scpu2	SCPU	R		data / Number	51 (°C)
/devicestatus/temperature/do1	SCPU	R		data / Number	25 (°C)
/devicestatus/temperature/do2	SCPU	R		data / Number	28(°C)
/devicestatus/voltage/us	SCPU	R		data / Number	24254 (mV)
/devicestatus/voltage/ua	SCPU	R		data / Number	25236 (mV)
/devicestatus/current/us	SCPU	R		data / Number	82 (mA)
/devicestatus/errorlog			logger	Structure	IOTC-4395 - The draft/logger profile
/devicestatus/errorlog/loglist	COM	S		Service	

Approved, Software, QM

5.3.1.5 Devicecontrol

SRIO-6561 - devicecontrol provides the following elements

tree element	Source of value	Rights / View	Profile	ElementType / DataType	Format (examples)
/devicecontrol			devicedetection		IOTC-592 - The devicereset profile
/devicecontrol/signal	COM	S		Service	triggers signalize (blinking RDY-LED)

Approved, Software, QM

5.3.1.6 Systemtime

SRIO-19921 - systemtime provides the following elements

tree element	Source of value	Rights / View	Profile	ElementType / DataType	Format (examples)
/systemtime			systemtime		IOTC-3235 - The systemtime profile
/systemtime/systick	COM	R		data / Number	returns the current systick / uptime. The value is in 'ms'

Approved, Software, QM

5.3.1.7 DI Elements

SRIO-2945 - Elements: DI Port

For each port the following elements shall exist:

tree element	Source of value	Rights / View	Profile	ElementType / DataType	Format (examples)
/io				Structure	
/io/di/port[n]				structure	n = 1..6
/io/di/port[n]/pin2				structure	
/io/di/port[n]/pin2/digital_input	COM	R		data / Boolean	1: Pin2 High 0: Pin2 Low
/io/di/port[n]/pin2/qualifier	COM	R		data / boolean	1: Input state good 0: Input state bad
/io/di/port[n]/pin4				structure	
/io/di/port[n]/pin4/digital_input	COM	R		data / boolean	1: Pin4 High 0: Pin4 Low
/io/di/port[n]/pin4/qualifier	COM	R		data / boolean	1: Input state good 0: Input state bad

Approved, Software, QM

5.3.1.8 DO Elements

SRIO-7996 - Elements: DO Port

For each port the following elements shall exist:

tree element	Source of value	Rights / View	Profile	ElementType / DataType	Format (examples)
/io				Structure	
/io/do/port[n]				Structure	n = 7..8
/io/do/port[n]/current_measurement	SCPU	R		data / Number	82 (mA)
/io/do/port[n]/pin2				Structure	
/io/do/port[n]/pin2/digital_output	COM	R		data / Boolean	1: Pin2 High 0: Pin2 Low
/io/do/port[n]/pin2/qualifier	COM	R		data / Number	1: Output state good 0: Output state bad
/io/do/port[n]/pin4				Structure	
/io/do/port[n]/pin4/digital_output	COM	R		data / Boolean	1: Pin4 High 0: Pin4 Low
/io/do/port[n]/pin4/qualifier	COM	R		data / Number	1: Output state good 0: Output state bad

Approved, Software, QM

5.3.1.9 Fieldbussetup

SRIO-2948 - Fieldbussetup provides the following elements:

tree element	Source of value	Rights / View	Profile	ElementType / DataType	Format (examples)
/fieldbussetup				Structure	
/fieldbussetup/fieldbusfirmware	COM	R		data / String	"3.3.0.12 (PROFIsafe IO Device)"
/fieldbussetup/connectionstatus	COM	R		data / Enum	0: Disconnected 1: Connected
/fieldbussetup/network			network	Structure	
/fieldbussetup/network/macaddress	COM	R		data / String	"00:02:01:03:F0:C5"
/fieldbussetup/network/ipaddress	COM	R		data / String	"172.18.87.2"
/fieldbussetup/network/subnetmask	COM	R		data / String	"255.255.192.0"
/fieldbussetup/network/ipdefaultgateway	COM	R		data / String	"172.18.64.1"
/fieldbussetup/network/hostname	COM	R		data / String	"fieldbusdevice1"
/fieldbussetup/network/dhcp	COM	R		data / Enum	0: Static IP

Approved, Software, QM

5.3.1.10 SafeCom elements

SRIO-2952 - For the profisafe parameter following elements are provided:

tree element	Source of value	Rights / View	Format (examples)
/safecom		Structure	
/safecom/f_address	SCPU	R	899

Approved, Software, QM

5.3.1.11 Firmware

SRIO-2953 - Firmware provide the following elements:

tree element	Source of value	Rights / View	Profile	ElementType / DataType	Format (examples)
/firmware			software, software/uploadablesoftware	Structure	IOTC-559 - The software profile Profile IOTC-566 - The software/uploadablesoftware profile
/firmware/version		R		data / String	UpdateContainer FW
/firmware/type	COM	R		data / String	"firmware"
/firmware/install		S [U]		Service	Initiades a reboot to install the firmware after an successfully upload
/firmware/container		W	blob	Structure	Firmware update via BLOB IOTC-568 - The blob profile
/firmware/container/maxsize	COM	R		data / Number	Maximum size of firmware blob to upload (hardcoded) 4128768 byte
/firmware/container/chunksize	COM	R		data / Number	Size of chunk for upload (hardcoded) 2048 byte
/firmware/container/start_stream_set		S [U]		Service	Start firmware upload.
/firmware/container/stream_set		S [U]		Service	Send firmware upload chunk.

Approved, SystemArchitecture, Software, QM

5.3.1.12 FIT

SRIO-17130 - FIT

tree element	Source of value	Rights / View	Profile	ElementType / DataType	Format (examples)
/fit				Structure	
/fit/setfit	COM	WO (P)		Service	

Approved, Software, QM

SRIO-17129 - Setfit

The service shall trigger a FIT for the device.

The available types are here:
40_505_Fault_Injection_Tests

Request parameters

Parameter	M/O/C	DataType	Description
type	M	Number	Identifies the fit test, that shall be executed.

Response parameter

None

Approved, Software, QM

6 Profinet

SRIO-1529 - PROFINET - Fieldbus Protocol PROFINET
The product must be able to perform the fieldbus protocol PROFINET in the role of a PROFINET IO Device. Supported PROFINET version: 2.4MU3 or higher.

Note: The supported version depends on which PROFINET version is supported by the 3rd-party PROFINET solution (e.g. netX90 PROFINET).

Approved, Software, PQS, QM

The building block delivers for software support:

- a driver for fieldbus chip with an example for peripheral unit
- an abstraction layer for fieldbus chip
- example fieldbus application
- example fieldbus description file
- production scripts
- update functions for fieldbus firmware (stack)

Details are described on gitlab: https://vhip.pages-ee.dev.ifm/fieldbus/documentation/Home/index.html

6.1 Features

6.1.1 NetX90

SRIO-3031 - NetX90 --- netIdent Protocol
The netIdent protocol shall be disabled.

Approved, Software, QM

6.1.2 DCP

SRIO-7772 - DCP --- Factory Reset
The DCP-reset service reset all stored data to its factory default values.
This covers all data related to application (Manufacturer specific record data and all I&M data) and communication parameters (NameOfStation shall be set to "" (empty string) and IP suite parameter shall be set to 0.0.0.0).

Remark:
Factory reset exists on different level, Mode 2 and 8 shall be tested.

Mode 1 - optional	Reset Application data	Resets I&M data, SNMP writeable OIDs, and manufacturer specific record data objects
Mode 2 - mandatory (default)	Reset Communication parameter	Resets name of station "", IP to 0.0.0.0, DHCP (if available), all PDev parameters (topology, etc.), and SNMP writeable OIDs
Mode 3 - optional	Reset Engineering parameter	Performs a Mode 1,2 reset + clears any engineering data (ex: configuration/parameters)
Mode 4 - optional	Reset All stored data	Performs a Mode 1, 2, & 3 reset
Mode 8 - optional	Reset device	Performs a Mode 4 reset to all interfaces in case the device has multiple interfaces
Mode 9 - optional	Reset and Restore Data	Same as mode 8 + resets installed software images back to factory default
Factory Reset (original default) - recommended	Original Factory Reset	Performs mode 1/2 reset, but can also could do mode 3,4,8, or 9 depending on the device behavior

Approved, Software, PQS, QM

SRIO-3061 - DCP --- Identity
To identify the device by reception of a DCP set Signal, the SF/BF LED (red) shall switch to blinking mode.

Approved, Software, QM

SRIO-3029 - DCP --- Setting Parameters
A PROFINET IO-Controller or an Engineering System can change IP parameters or the NameOfStation of device at any time.

Approved, Software, QM

6.1.3 [IRT]

SRIO-3025 - IRT
The safety Remote IO shall support isochronous Real time functionality (IRT).
But no submodule supports isochronous mode, i.e process data exchange must not be synchronous to RTsync messages.

Remark: SRIO shall not disturb IRT Communication, also this results in an additional test for certification.

Approved, Software, QM

6.1.4 MediaRedundancyProtocol

SRIO-1930 - PROFINET - MRP Support
The product shall support the Media Reduncancy Protocol in the role as Media Redundancy Client.

Approved, Software, QM

6.1.5 Fieldbus Interface

SRIO-3027 - Port Deactivation
The safety Remote IO shall support port deactivation on both fieldbus ports.

Approved, Software, QM

SRIO-10247 - Port Deactivation
It shall not be possible for the user to deactivate both ports at the same time.

Approved, Software, QM

6.1.6 Supported Profile

SRIO-3028 - Profinet Profile - RIOforFA with PROFIsafe
The safety Remote IO shall support the PROFINET Profile "Remote IO for Factory Automation with PROFIsafe", RIOforFA.
The RIOforFA Profile fulfills the following requirements:
- Define Qualifier to Values of Physical Signals, The Qualifier shall be synchronous and consistent with the Value
- Define a RIOforFA channel for digital IO. One RIOforFA channel consists of Value and Qualifier.
- The type and quantity/number of RIOforFA channels are defined by GSD.
- A RIOforFA (Sub)Module uses RIOforFA channels only.
- The safety device uses PROFIsafe standard. This Profile does not add any definition to standard diagnosis (PROFINET Diagnosis Guideline) and I&M Functions.

Remark: This requirement is achieved by one Submodule, while the other will provide legacy support.

Approved, Software, QM

6.2 GSD File

6.2.1 General

SRIO-1915 - PROFINET - Conformance Class C (CC-C)
The product shall fulfill Conformance Class C (CC-C).

Approved, Software, QM

SRIO-1916 - PROFINET - Net Load Class III
The product shall fulfill Net Load Class III.

Approved, Software, QM

SRIO-3019 - GSD --- File Name
The GSD filename is GSDML-V2.yy-ifm-AL40xS-xxxxxxxx.xml
where xxxxxxxx is the date of creation of GSD file (e.g. 20220210) and yy the GSDML version used.

Approved, Software, QM

SRIO-3018 - GSD --- Language of GSDML File
Language of GSDML File is english only. Approved, Software, QM

SRIO-2965 - GSD --- Device IDs
The SRIO shall support the following Device IDs:

Devices	Vendor ID	DeviceID
AL400S, AL401S	0x0136	**0xAC25**

Table 3 Devices IDs Approved, Software, QM

SRIO-3020 - GSD --- Device Identity
The SRIO shall have the following identity parameter:

GSD Parameter	Value
VendorID	0x0136
Vendor	"ifm electronic"
InfoText	"ETH Module PROFIsafe 6x2 F-DI 2x2 F-DO"

Table *4* : Device Identity

Approved, Software, QM

SRIO-3021 - GSD --- Device Function
The SRIO shall be part of the following product family:

GSD Parameter	Value
ProductFamily	"Digital I/O modules"
MainFamily	"I/O"

Table 5 : Device Function

Approved, Software, QM

SRIO-1864 - GSD --- Backwards compatibility
The GSDML scheme shall be backwards compatible to *Technical Specification for PROFINET: GSDML, Version 2.31, 2016.*
Approved, Software, QM

SRIO-3017 - GSD --- Process data length
The Maximum lengths are defined by maximum process data of all submodules and IOXS bytes of all subslots.

Device	Maximum Input Length [Bytes]	Maximum Output Length [Bytes]
AL400S, AL401S	17	15

Approved, Software, PQS, QM

SRIO-3030 - GSD --- GSDML attribute "ResetToFactoryModes"
GSDML attribute "ResetToFactoryModes" shall be set to "2".
This mode is intended to set the addressed communication interface of a device into a state, which is almost similar to the "out of the box" state.
In particular these are:
- NameOfStation shall be set to "" (empty string)
- IP suite parameter shall be set to 0.0.0.0

Remark: I&M data are not set to factory values Approved, Software, QM

SRIO-7758 - GSD --- CRC Signature
The GSDML attribute "F_ParamDescCRC" [CRC Signature] shall be calculated according to the PROFIsafe specification and set as value in the GSDML.
Approved, Software, PQS, QM

6.2.2 F-Parameters

<u>Address Type</u>
Address Type 1 if F_DestAdd is checked only

Address Type 2 if F_SourceAdd is checked together with the F_DestAdd

SRIO-7830 - GSD --- FPar --- Address Type

Safe Remote IO shall support Address Type 1. Approved, Software, PQS, QM

SRIO-7794 - GSD --- FPar --- F-Source Address

The F-Source Address shall be set in range [1...65534] via Fieldbus/safe fieldbus engineering tool.

Remark: In Address Type 1, the F-Source address will be not checked. Approved, Software, PQS, QM

The F-Destination Address is the own safe address of the safe Remote IO. The configured F-Destination-Address via Fieldbus/safe fieldbus engineering tool shall be the same as the value set by the rotary switches.

SRIO-1857 - GSD --- FPar --- Selectable F_WD_Time

F_WD_Time of the F-Parameters shall be selectable between minimal F_Watchdog_Time and 10000 ms with the provided GSDML.

Note: The minimal F_Watchdog_Time is specified in SRIO-1210 - Timings - F_WD_Time Parameter.

Approved, Software, SIL 3

SRIO-2999 - GSD --- FPar --- F-Parameters Index

For the F-Parameters, the Index **0x200** shall be assigned

Approved, Software, QM

SRIO-2998 - GSD --- FPar --- F-Parameter record data

Byte	Parameter	Data Type	Description	Valuation / Value	Changeable
1	F_Prm_Flag1	Bit 0	F_Check_SeqNr	V2-mode: to be ignored	no
		Bit 1	F_Check_iPar	Manufacturer specific use: to be ignored	no
		Bit 2 and Bit 3	F_SIL: configured SIL	SIL3	no
		Bit 4 and Bit 5	F_CRC_Length: describes the length of the CRC2 signature	for ProfiSafe Version >= 2.6, F_CRC_Length = "4-Byte-CRC"	no
		Bit 6	F_CRC_Seed	"CRC-Seed24/32"	no
		Bit 7	reserved	-	
2	F_Prm_Flag2	Bit 0	F_Passivation: F-(Sub)Module or Channel-granular Passivation	Channel-granular Passivation	no
		Bit 1 and Bit 2	reserved	-	
		Bit 3 to Bit 5	F_Block_ID: Parameter block type identification (Bit 0 indicates the presence of i-Parameters (F_iPar_CRC). Bit 1 indicates the presence of the secondary F-Watchdog time)	1	no
		Bit 6 and Bit 7	F_Par_Version: Version number of F-Parameter block	1	no
3	F_Source_Add	UINT16	F-Address of the F-Host	[1...65534] (2^16)-1	Yes Hint: The Source Address must

					be changeable. Req. by GSD-Checker.
5	F_Dest_Add	UINT16	The own F-Address shall be configurable via the rotary switches in the range [1..899]	[1..899], Set to default value 1	Yes
7	F_WD_Time	UINT16	Watchdog time in the F driver	SRIO-1857 - GSD --- FPar --- Selectable F_WD_Time [50...10000ms]	Yes
9	F_iParCRC	UINT32	Value of the iParameter CRC signature calculation, at least manually transferred from a CPD-Tool to the engineering tool	[0..4294967295] (2^32)-1	Yes
13	F_ParCRC	UINT16	Signature calculation across the F-Parameters to secure the transfer from the F-Engineering Tool to the F-Device	[0..65535] (2^16)-1	Yes

Table 6 F-Parameters

Approved, Software, QM

6.2.3 Individual Parameter of Safety Digital Input

SRIO-2989 - GSD --- iPar --- Record Indexes of safety digital Input

For each safety digital input an index in the range **0x30n** should be assigned.
FDI1(Safety Digital Input 1) starts with **0x301**

Approved, Software, QM

SRIO-2990 - Parameters of safety Digital Input:

The following Parameter for one DI Port shall apply:
(**bold** text shall be default)

Byte	Parameter	Data type	Description of Parameter	Valuation
0	F-DI 1	UINT8	Enables the first input of the respective port	0 : **Disabled** 1 : Enabled
1	F-DI 2	UINT8	Enables the second input of the respective port.	0 : **Disabled** 1 : Enabled
2	Testpulse Supply 1	UINT8	Supply Channel 1	0 : Disabled 1 : **Enabled**
3	Testpulse Supply 2	UINT8	Supply Channel 2	0 : Disabled 1 **: Enabled**
4	Symmetry	UINT8	Defines if the signals are processed individually (1oo1) or in combination to each other (1oo2)	0: 1oo1 **1: 1oo2 equivalent** 2: 1oo2 antivalent
5	Symmetry Discrepancy Time	UINT8	In case of symmetry, the symmetry discrepancy time setting defines the timespan in which the symmetry definition shall be achieved.	**0: Not Analysed** 1: 10 ms, 2: 20 ms, 3: 50 ms, 4: 100 ms, 5: 500 ms, 6: 1000 ms, 7: 5000 ms,
6	Resolve Symmetry Violation	UINT8	Defines, in case of a symmetry violation, if a test-0-signal shall be necessary or if an	0: **Acknowledgment sufficient** 1: Switch-on inhibit

			acknowledgment is sufficient.	
7	Filter F-DI 1	UINT8	Filter Time Channel 1	1: 5ms, **2: 10ms**
8	Filter F-DI 2	UINT8	Filter Time Channel 2	1: 5ms, **2: 10ms**

Table 7 Parameters Safe Input

Approved, Software, QM

6.2.4 Individual Parameters of Safety Digital Output

SRIO-3001 - GSD --- iPar --- Record Indexes of safety digital Output
For each safety digital output an index in the range **0x31n** should be assigned.
FDO1(Safety Digital Output 1) starts with **0x311**

Approved, Software, QM

SRIO-3000 - Parameter of safety digital Output
The following Parameter for one DO Port shall apply:
(**bold** text shall be default)

Byte	**Parameter**	**Data Type**	**Description of Parameter**	**Parameter-Range**
0	F-DO 1	UINT8	Enables the first output of the port.	0 : **Disabled** 1 **:** Enabled
1	F-DO 2	UINT8	Enables the second output of the port.	0 : **Disabled** 1 **:** Enabled
2	Testpulse	UINT8	Defines if the testpulse shall be activated.	0 : Disabled **1 : Enabled**
3	Testpulse Duration	UINT8	Defines the testpulse duration [fix]. *(available for user information)*	*[0 reserved for future 0,1 ms]* 1 : 0,5 ms **2 : 1 ms** 3 : 3 ms
4	Symmetry	UINT8	Defines if the signals are processed individually (1oo1) or in combination to each other (1oo2).	0: 1oo1 P **1: 1oo2 PP** 2: 1oo2 PM
5	Switch-off Delay F-DO 1	UINT8	Defines a switch-off Delay of the first output of the port	**0: Disabled,** 1: 250 ms, 2: 500 ms, 3: 1000 ms, 4: 2000 ms, 5: 4000 ms, 6: 8000 ms
6	Switch-off Delay F-DO 2	UINT8	Defines a switch-off Delay of the second output of the port.	**0: Disabled**, 1: 250 ms, 2: 500 ms, 3: 1000 ms, 4: 2000 ms, 5: 4000 ms, 6: 8000 ms

Table 8 Parameters Safe Output

Approved, Software, QM

6.2.5 Modules and Submodules IDs

The GSDML consists of Device Access Points [DAP] (one for each variant), Modules and Submodules

SRIO-6563 - GSD --- Modular Concept

The device GSD shall group functionalities in modules and submodules. Approved, Software, PQS, FSM, QM

SRIO-14859 - GSD --- Module Sensor Reactivation

The device GSD shall provide a module "Sensor Reactivation" compatible to both DAPs.

Approved, Software, PQS, FSM, QM

SRIO-14858 - GSD --- Module PROFIsafe Channel granular

The device GSD shall provide a PROFIsafe Module, with "Channel granular" Passivation, compatible to both DAPs.

Approved, Software, PQS, FSM, SIL 3

SRIO-1931 - GSD --- PROFINET Profile "Remote IO Factory Automation"

The PROFIsafe Module with "Channel granular Passivation" shall support the PROFINET Profile "Remote IO Factory Automation", Version 1.10, August 2018. Approved, Software, PQS, FSM, QM

SRIO-14857 - GSD --- Module PROFIsafe Device granular

The device GSD shall provide a PROFIsafe Module, with Passivation "Device", compatible to both DAPs.

Approved, Software, PQS, FSM, SIL 3

SRIO-14856 - GSD --- DAP Headmodule

The device GSD shall provide two DAPs for each AL400S and AL401S. Approved, Software, PQS, FSM, QM

SRIO-2973 - GSD --- Profinet Module IDs

The Profinet Module should have following IDs:

ID	Description	Profile ID
0x0000 4000	DAP Module AL400S	0x0000
0x0000 4010	DAP Module AL401S	0x0000
0x1000 0000	MI_SENSOR_REACTIVATION_ID	0x0000
0x2000 0000	MI_6FDI2FDO_CHANNEL_PASSIVATION_ID	RioForFA = 0x4C00
0x3000 0000	MI_6FDI2FDO_MODULE_PASSIVATION_ID	0x0000

Approved, Software, QM

SRIO-2975 - GSD --- Profinet (Virtual-)Submodule IDs

The Profinet (Virtual-)Submodule should have following IDs:

ID	Description
0x0001 0000	DAP Virtual Submodule Item
0x0001 0010	Interface Submodule Item
0x0001 0011	Port Submodule Item X1 P1
0x0001 0012	Port Submodule Item X1 P2
0x10000100	Sensor Reactivation Module
0x2000060A	6DI/2DO Submodule - Channel granular Passivation
0x3000060A	6DI/2DO Submodule - Module Passivation

Approved, Software, QM

SRIO-6552 - GSD --- PROFIsafe - F_Parameter Module --- PROFIsafe Channel granular

The "*6DI/2DO Submodule - Channel granular Passivation*" shall support the following F-Parameter settings:

Parameter-Name	**Value range**	**Changeable**	**Description**
F_SIL	SIL3	no	
F_CRC_Length	4-Byte-CRC	no	
F_Block_ID	1	no	

F_Par_Version	1	no	
F_Source_Add	1...65534	yes	(Has to be changeable to pass GSD-Checker)
F_Dest_Add	1...899	yes	
F_Passivation	**Channel**	no	
F_CRC_Seed	CRC-Seed24/32	no	
F_WD_Time	50...10000ms default: **150ms**	yes	PROFIsafe Watchdog Time; Lowest value to be defined in implementation.
F_iPar_CRC	0...FFFFFFFF	yes	
F_Par_CRC	0...65535	yes	

Approved, Software, PQS, FSM, SIL 3

SRIO-14860 - GSD --- PROFIsafe - F_Parameter Module --- PROFIsafe Device granular
The "*6DI/2DO Submodule - Module Passivation*" shall support the following F-Parameter settings:

Parameter-Name	**Value range**	**Changeable**	**Description**
F_SIL	SIL3	no	
F_CRC_Length	4-Byte-CRC	no	
F_Block_ID	1	no	
F_Par_Version	1	no	
F_Source_Add	1...65534	yes	(Has to be changeable to pass GSD-Checker)
F_Dest_Add	1...899	yes	
F_Passivation	**Device**	no	
F_CRC_Seed	CRC-Seed24/32	no	
F_WD_Time	50...10000ms Default: **150ms**	yes	PROFIsafe Watchdog Time; Lowest value to be defined in implementation.
F_iPar_CRC	0...FFFFFFFF	yes	
F_Par_CRC	0...65535	yes	

Approved, Software, PQS, FSM, SIL 3

6.3 I&M Data

The I&M functions are subdivided into different blocks, which can be addressed separately. Each Profinet device must provide at least one submodule as a representative of the device that supports the I&M0-I&M3 functions. Additionally a PROFIsafe field device should should support the I&M4 function.

Usage	**IM_INDEX**	**Access**
I&M0	0xAFF0	Read
I&M1	0xAFF1	Read / Write
I&M2	0xAFF2	Read / Write
I&M3	0xAFF3	Read / Write
I&M4	0xAFF4	Read
I&M5	0xAFF5	Read

6.3.1.1 I&M0 Filter Data

This is a special read-only record (0xF840) identifying which submodules carry distinct I&M data. More than one submodule may be specified by repeating this structure. When reading it, addressing of the slot/subslot is ignored.
For the submodule acting as device representive I&M1, I&M2 and I&M3 shall be readable and writable. Usually the representive for the device is the Interface or the DAP

SRIO-3015 - I&M --- I&M0 Filter Data

Variable	Bytes	Value	Description
API	4		API of the submodule
SLOT	2		Slot of the submodule
SUBSLOT	2		Subslot of the submodule
FLAGS	4		Flags: DAP: 0x01 submodule has own I&M data; 0x04 submodule represents device I&M data Sensor Reactivation submodule: 0x02 submodule has I&M data Safety RIO submodule: 0x02 submodule has I&M data

Approved, Software, PQS, QM

6.3.1.2 I&M0

SRIO-3002 - I&M --- I&M0
I&M0 shall be readabe on every submodule

Variable	Bytes	Value	Description
MANUFACTURER_ID	2	"136_{hex}"	Vendor ID
ORDER_ID	20	AL40xS	OrderID (ASCII, padded with spaces)
SERIAL_NUMBER	16		SerialNumber (ASCII, padded with spaces)
HARDWARE_REVISION	2	e.g. AA	HardwareRevision
SOFTWARE_REVISION	4	e.g. V1.0.2	<table><tr><th>Byte</th><th>Description</th></tr><tr><td>0</td><td>Software-Typ (char): V : Release</td></tr><tr><td>1</td><td>Major Version (uint8)</td></tr><tr><td>2</td><td>Minor Version (uint8)</td></tr><tr><td>3</td><td>Build Version (uint8)</td></tr></table>Build number is our branch version
REVISION_COUNTER	2	1...0xFFFF	Starting from 0, shall increment on each parameter change.
PROFILE_ID	2	**0x0000** (for andere Submodule) **0x4C00** (for Safe Submodule)	The profile of the (sub) module. (see https://www.profibus.com/IM/Profile_ID_Table.xml)

PROFILE_SPECIFIC_TYPE	2	0x0000	ProfileSpecificType (additional value to ProfileID) not used 0
IM_VERSION	2	0x0101	IMVersion (I&M version, default 0x0101)
IM_SUPPORTED	2	**0x000E** for DAP **0x0030** for Safety Submodule **0x0000** For Sensor reactivation Submodule	0x2E : **I&M0,1,2,3**supported 0x0010: **I&M4 and 5** supported 0x0000 **I&M0** supported

Approved, Software, QM

6.3.1.3 I&M1

SRIO-7769 - I&M --- I&M1 availability
I&M1 shall be only available for the DAP.
Otherwise attempting to read or write I&M1 will result in an invalid index error. Approved, Software, QM

SRIO-7768 - I&M --- I&M1 Access
I&M1 is readable and writable. Approved, Software, QM

SRIO-7767 - I&M --- I&M1 Storage
I&M1 shall be stored remanent (powercycle proof). Approved, Software, QM

SRIO-3005 - I&M --- I&M1

Variable	Bytes	Value (Default)	Description
TAG_FUNCTION	32	"20_{hex}" (empty)	Tag Function of submodule (ASCII, padded with spaces)
TAG_LOCATION	22	"20_{hex}" (empty)	Tag Location of submodule (ASCII, padded with spaces)

Approved, Software, QM

SRIO-7770 - I&M --- I&M1 on factory reset
On factory reset, I&M1 shall be reset to a empty string (54 x 20hex). Approved, Software, QM

6.3.1.4 I&M2

SRIO-7766 - I&M --- I&M2 availability
I&M2 shall be only available for the DAP.
Otherwise attempting to read or write I&M2 will result in an invalid index error. Approved, Software, QM

SRIO-7765 - I&M --- I&M2 Access
I&M2 is readable and writable Approved, Software, QM

SRIO-7764 - I&M --- I&M2 Storage
I&M2 shall be stored permanently (powercycle proof). Approved, Software, QM

SRIO-3004 - I&M --- I&M2

Variable	Bytes	Value (Default)	Description
INSTALLATION_DATE	16	"20_{hex}" (empty)	Installation Date of submodule (ASCII, padded with spaces)
RESERVED	38	00_{hex}	Reserved, set to zero

Approved, Software, QM

SRIO-7763 - I&M --- I&M2 on factory reset
On factory reset, I&M2 shall be reset to a empty string (16 x 20hex). Approved, Software, QM

6.3.1.5 I&M3

SRIO-7759 - I&M --- I&M3 availability
I&M3 shall be only available for the DAP.

Otherwise attempting to read or write I&M3 will result in an invalid index error. Approved, Software, QM

SRIO-7760 - I&M --- I&M3 Access
I&M3 is readable and writable. Approved, Software, QM

SRIO-7761 - I&M --- I&M3 Storage
I&M3 shall be stored permanently (powercycle proof). Approved, Software, QM

SRIO-3003 - I&M --- I&M3

Variable	Bytes	Value (Default)	Description
DESCRIPTOR	54	"20_{hex}" (empty)	Descriptor of submodule (ASCII, padded with spaces)

Approved, Software, QM

SRIO-7762 - I&M --- I&M3 on factory reset
On factory reset, I&M3 shall be reset to a empty string (54 x 20hex). Approved, Software, QM

6.3.1.6 I&M4

SRIO-7700 - I&M --- GSDML-Check:
The DAPs shall not offer write access to I&M4. Approved, Software, PQS, QM

SRIO-7705 - I&M --- I&M4 - Supported Module
I&M4 data is only supported by the PROFIsafe Modules. Approved, Software, QM

SRIO-7706 - I&M --- Read/Write Access of I&M4
I&M4 data shall only be readable (not writable). Approved, Software, QM

SRIO-7704 - I&M --- I&M4 Storage
The I&M4 Data is generated during the runtime, and is not stored remanently. Approved, Software, QM

SRIO-3011 - I&M --- I&M4 data
The I&M4 Data shall be a string containing the value of F_iParCRC and the F_Par_CRC from the F-parameter

Variable	Bytes	Value (Default)	Description
SIGNATURE	54	"0x00"	Signature generated by application Byte 0 - 3 : F_iParCRC (Value in Hex) Byte 4 - 6 : F_Par_CRC (Value in Hex)

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6.3.1.7 I&M5

SRIO-7707 - I&M --- Read/Write Access of I&M5
I&M5 shall be only readable. Approved, Software, QM

SRIO-3023 - I&M --- I&M5

Variable	Bytes	Value (Default)	Description
ANNOTATION	64	"host_fw_1.0.3.215"	Manufacturer specific annotation string Padded with spaces (0x20) to 64 byte length Firmware Version with Prefix e.g "host_fw_1.0.3.215"
ORDER_ID	64	"20_{hex}" (empty)	Order Id padded with spaces (0x20) to 64 byte length
VENDOR_ID	2		VendorId
SERIAL_NUMBER	16		Serial number Padded with spaces (0x20) to 16 byte length

HARDWARE_REVISION	2		Hardware revision
SOFTWARE_REVISION.PREFIX	1	'V'	Character describing the software of the OEM part. Allowed values: 'V', 'R', 'P', 'U' and 'T'
SOFTWARE_REVISION.X	1		Major version number
SOFTWARE_REVISION.Y	1		Minor version number
SOFTWARE_REVISION.Z	1		Build version number

Remark: The read FW-Version shall be the Version of the Host.

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6.4 Profinet Mapping

Channel Assignment

SRIO-2060 - DI/DO Numbering, DI/DO Device
For a DI/DO device the DI/DO numbers shall denote the port and port-channels as follows:

DI/DO number	Port	Port Pin	Function
DI 1	1	4	Channel 0
DI 2	1	2	Channel 1
DI 3	2	4	Channel 2
DI 4	2	2	Channel 3
DI 5	3	4	Channel 4
DI 6	3	2	Channel 5
DI 7	4	4	Channel 6
DI 8	4	2	Channel 7
DI 9	5	4	Channel 8
DI 10	5	2	Channel 9
DI 11	6	4	Channel 10
DI 12	6	2	Channel 11
DO 1	7	4	Channel 12
DO 2	7	2 (P/PP) or 3 (PM)	Channel 13
DO 3	8	4	Channel 14
DO 4	8	2 (P/PP) or 3 (PM)	Channel 15

Figure 4 : DI/DO numbers for 6x2 F-DI and 2x2 F-DO

Note: The Port Pinning is defined in SRIO-1990 - IO Connectors

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6.4.1 Cyclic Data

SRIO-2986 - The cyclic process data of 6DI/2DO module
The 6DI/2DO safety module shall have following inputs and outputs data mapping:

SRIO to F-Host/ PLC - Input Data (for F-Host/ PLC):

Byte/Bit	7	6	5	4	3	2	1	0

n	Digital Input Channel 07	Digital Input Channel 06	Digital Input Channel 05	Digital Input Channel 04	Digital Input Channel 03	Digital Input Channel 02	Digital Input Channel 01	Digital Input Channel 00
n + 1	not used	not used	not used	not used	Digital Input Channel 11	Digital Input Channel 10	Digital Input Channel 09	Digital Input Channel 08
n + 2	Qualifier for Digital Input Channel 07	Qualifier for Digital Input Channel 06	Qualifier for Digital Input Channel 05	Qualifier for Digital Input Channel 04	Qualifier for Digital Input Channel 03	Qualifier for Digital Input Channel 02	Qualifier for Digital Input Channel 01	Qualifier for Digital Input Channel 00
n + 3	not used	not used	not used	not used	Qualifier for Digital Input Channel 11	Qualifier for Digital Input Channel 10	Qualifier for Digital Input Channel 09	Qualifier for Digital Input Channel 08
n + 4	not used	not used	not used	not used	Qualifier for Digital Output Channel 15	Qualifier for Digital Output Channel 14	Qualifier for Digital Output Channel 13	Qualifier for Digital Output Channel 12
n + 5	PROFIsafe Status Byte							
n + 6... n + 9	PROFIsafe Checksum (CRC)							

F-Host/ PLC to SRIO - Output Data (from F-HOST/ PLC):

Byte/Bit	7	6	5	4	3	2	1	0
n	not used	not used	not used	not used	Digital Output Channel 15	Digital Output Channel 14	Digital Output Channel 13	Digital Output Channel 12
n + 1	PROFIsafe Control Byte							
n + 2... n + 5	PROFIsafe Checksum (CRC)							

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SRIO-7354 - Power Switch module for 6DI/2DO device

The Power Switch module shall have following mapping **[Output Data]:**

Byte/Bit	7	6	5	4	3	2	1	0
0	Power switch input channel 7	Power switch input channel 6	Power switch input channel 5	Power switchinput channel 4	Power switch input channel 3	Power switch input channel 2	Power switch input channel 1	Power switch input channel 0
1	not used	not used	not used	not used	Power switch input channel 11	Power switch input channel 10	Power switch input channel 9	Power switch input channel 8

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6.5 Acyclic Access

SRIO-7749 - Acyclic write access after config phase

After the param-end command, a write to the configuration records shall be blocked.

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6.6 Diagnosis and Process alarms

6.6.1.1 Process Alarm

SRIO-7817 - Process alarm

The safe remote IO shall not support process alarms. Approved, Software, PQS, QM

6.6.1.2 Diagnosis alarm

6.6.1.2.1 Channel Number

SRIO-7823 - Channel Number

For all diagnosis alarms the source of the diagnosis shall correspond to the channel.

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6.6.1.3 Channel proprieties

In the channel properties, the application specifies data such as the following:

- Maintenance is required or demanded
- Diagnostic alarm is comming in/going out

SRIO-7822 - Channel property

For all diagnosis, "Diagnosis" shall be used as channel property.

Channel property = 0x0000 Approved, Software, PQS, QM

6.6.1.4 Channel Error type

The device specifies the cause of an error in the data area 'ChannelErrorType'.

SRIO-7825 - Channel Error Type

All application specific diagnosis alarms shall be defined in the manufacturer specific range 0x100 - 0x7FFF.

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